The COVID-19 Shock to Primary Education: Cross-Cohort Evidence from Ukraine's State Monitoring Programme

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March 4, 2026

Abstract

This paper estimates COVID-19-related learning losses among Ukrainian primary students using two waves of nationally representative assessments administered before (May 2018) and after (May 2021) extended school closures. Average mathematics scores declined by 0.09 standard deviations after controlling for region, school type, and student socioeconomic background, equivalent to roughly one-fifth of a typical school year. Reading scores declined by a similar magnitude (0.09 SD). These losses occurred despite substantial variation in remote-instruction practices: nearly half of students experienced 1–2 months of distance learning, while 13 percent were exposed for three months or longer. The findings contribute scarce evidence on pandemic-related educational disruption in middle-income countries and suggest that Ukraine's primary students experienced learning setbacks comparable to those documented in other national contexts.

Keywords: COVID-19, learning loss, primary education, Ukraine; **JEL:** I21, I24

1 Introduction

The COVID-19 pandemic disrupted schooling worldwide through prolonged closures and sharp reductions in instructional time. According to UNESCO, more than 1.6 billion learners across over 190 countries experienced school interruptions during the pandemic (UNESCO, 2021). A growing body of empirical research suggests that such disruptions can translate into persistent losses in human capital, particularly among disadvantaged students.

Recent systematic reviews quantify both the magnitude of pandemic-related learning losses and their unequal distribution. Betthäuser et al. (2023), synthesizing evidence from 42 studies across 15 countries, document an average learning deficit of approximately 0.14

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standard deviations (SD), equivalent to roughly one third of a typical school year. Losses are consistently larger among socio-economically disadvantaged children, indicating that the pandemic exacerbated pre-existing educational inequalities. Reviewing 36 rigorous evaluations, Patrinos et al. (2022) report a mean shortfall of about 0.17 SD—around half a grade level—along with systematic widening of socio-economic gaps. Extending the evidence base to 56 studies, primarily from developing countries, Dela Cruz et al. (2024) estimate that each full year of school closure translates into approximately 1.1 years of lost learning and highlight the effectiveness of remediation policies such as tutoring delivered in person or by phone. Taken together, these reviews indicate that COVID-19 generated large, unequal, and only partially mitigated learning losses worldwide.

This paper contributes to studies that measure pandemic-related learning losses using in-person assessments administered to large samples of students (e.g., Hevia et al., 2022; Alasino et al., 2024; Ardington et al., 2021; Guariso and Björkman Nyqvist, 2023; Lichand et al., 2022; Singh et al., 2024). Ukraine provides a particularly informative setting as an upper-middle-income country for which nationally representative evidence on primary-school learning during the pandemic remains scarce. I use two rounds of the State Monitoring of Quality of Primary Education, which assessed fourth-grade achievement in mathematics and reading before the pandemic (May 2018) and after extended periods of remote and hybrid schooling (May 2021).

The paper contributes in three ways. First, it provides evidence from Ukraine using a nationally representative monitoring programme with standardized assessment administration and a common score scale across waves. Second, because the analysis compares cross-sections, I document sample comparability across waves and explicitly assess sensitivity to differential test participation and compositional change. Third, the monitoring programme links student test scores to classroom-level teacher questionnaires, which I use to document heterogeneity in distance-learning exposure and instructional practices during remote periods.

2 Background

2.1 Institutional Background

Ukraine is an upper-middle-income country. Primary education lasts four years (grades 1–4), followed by basic secondary education (grades 5–9) and upper secondary education. While the Ministry of Education and Science sets national curricula and standards, administration and financing are shared with sub-national governments, whose role expanded following decentralization reforms. Public education spending averaged between 4 and 6 percent of GDP in the years preceding the full-scale invasion (UNESCO, 2022).

Prior to the COVID-19 pandemic, educational access in Ukraine was nearly universal, with primary enrollment rates exceeding 95 percent. However, concerns about learning quality persisted. In Ukraine's first participation in PISA in 2018, 15-year-old students scored below the

OECD average in reading, mathematics, and science, highlighting challenges related to both overall performance and equity. Substantial disparities existed between urban and rural schools and across socio-economic groups, reflecting differences in resources and learning environments.

2.2 COVID-19 Disruption

All schools in Ukraine closed nationwide on 12 March 2020 and remained fully remote until the end of the academic year on 29 May 2020. Schools reopened in September 2020 under an adaptive-quarantine system, whereby in-person instruction was permitted in districts classified as green, yellow, or orange, but automatically suspended whenever a district entered the red zone. Because district classifications were updated frequently in response to local epidemiological conditions, exposure to remote instruction varied substantially across regions and over time.

According to UNESCO's global school-closure tracker, Ukrainian primary schools experienced approximately 18 weeks of full closure and 13 weeks of partial or hybrid operation between February 2020 and March 2022, equivalent to roughly 41 weeks (about 1.1 school years) of disrupted instruction (UNESCO, 2025). These aggregate figures reflect official closure policies rather than realized instructional exposure at the classroom level. In practice, the adaptive-quarantine regime generated considerable heterogeneity in how long students spent learning remotely.

Table 1 summarizes the timing of school-year start dates, monitoring cycles, and COVID-19 disruptions relative to the assessment periods used in this study.

Table 1: Timeline of school-year start dates, monitoring cycles, and COVID-19 disruptions in Ukraine

Date	Event
1 Sep 2017	Start of the 2017/18 school year (cohort tested in 2018)
17 Apr – 18 May 2018	First monitoring assessment (366 schools, 486 classes)
1 Sep 2019	Start of the 2019/20 school year (pandemic strikes mid-year)
12 Mar – 29 May 2020	Nationwide full shutdown (79 days; 11 weeks fully remote)
1 Sep 2020	Start of the 2020/21 school year under adaptive quarantine
8 Jan – 24 Jan 2021	Winter lockdown (17 days; 2.5 weeks fully remote)
25 Jan – 15 Apr 2021	Adaptive quarantine resumes
16 Apr – 17 May 2021	Second monitoring assessment (355 schools, 475 classes)

During the 2021 monitoring cycle, teachers reported the duration of remote instruction experienced by the assessed class over the 2020/2021 school year. Using these responses, I construct a class-level measure of realized exposure to distance learning. Table 2 reports the distribution of this measure for the full 2021 student sample.

Table 2: Distribution of 2021 students by teacher-reported duration of remote instruction

Duration of remote learning	Number of students	Share (%)
0 months	787	9.9
Less than 1 month	2,311	29.2
1–2 months	3,819	48.2
3–4 months	818	10.3
5 or more months	189	2.4

Nearly half of students experienced one to two months of remote instruction during the 2020/2021 school year, while approximately 13 percent were exposed for three months or more. At the other extreme, about 10 percent of students experienced no remote learning. This distribution highlights the gap between policy-level measures of school closures and realized classroom-level exposure, and provides a basis for examining heterogeneity in learning outcomes associated with differential exposure to distance education.

3 Data and Sample Characteristics

3.1 State Monitoring of Quality of Primary Education

I use two waves of Ukraine's State Monitoring of Quality of Primary Education, a nationally representative assessment programme designed to track achievement trends, evaluate alignment with national learning standards, and examine factors affecting student outcomes. The programme assesses fourth-grade students in mathematics and reading and collects parallel questionnaires from students and teachers.

The baseline sample was drawn from all Ukrainian-language general secondary schools with fourth-grade enrollment. Schools were stratified by size and selected using probability-proportional-to-size (PPS) sampling. The design yielded 366 schools and 486 classes, with an expected sample of 9,884 pupils. Each class was randomly assigned to complete either the mathematics or reading assessment. Fieldwork occurred 17 April–18 May 2018 and produced data on 9,077 tested pupils (4,535 in reading; 4,542 in mathematics). After standard exclusions for missing data, the analytic sample comprises 9,007 valid pupil records. All 486 sampled teachers completed questionnaires.

The 2021 wave followed identical sampling and administration procedures. The initial sample included 365 schools, 485 classes, and 9,241 pupils. Pandemic-related logistical challenges reduced the final achieved sample to 355 schools and 475 classes. Fieldwork occurred 16 April–17 May 2021, yielding data on 7,991 tested pupils (4,192 reading; 3,799 mathematics), of which 7,924 records are retained after exclusions. A total of 473 teachers completed questionnaires.

In both waves, assessments were administered under standardized protocols by external proctors. Materials were handled under secure conditions, and each class completed only one

subject test along with a student questionnaire.

Student achievement was scaled using item response theory (IRT). Dichotomous items were modeled using a two-parameter logistic (2PL) model, while polytomous items employed a graded response model. All item responses were placed on a common logit scale, ensuring comparability across rotated test forms and between waves. Logit scores were linearly transformed to a reporting scale of 100–300 points, with cut-scores at 170, 200, and 230 defining four performance levels: below-basic, basic, middle, and high. Because both waves use the same IRT parameters and reporting transformation, scores are directly comparable at the population level. Further technical details are available in programme documentation (UCEQA, 2025).

All analyses use survey sampling weights to recover population-representative estimates and cluster standard errors at the school level (primary sampling unit) to account for design effects and within-school correlation.

3.2 Sample Comparability and Attrition

Because this design compares repeated cross-sections rather than tracking the same students or schools, assessing comparability across waves is essential. Table 3 presents balance statistics for the mathematics sample.

Several patterns merit attention. First, enrollment declined by about 550 students (11%), and the proportion tested fell from 90.8 to 85.8 percent, losing approximately 720 tested students between waves. If lower-performing or more disadvantaged students differentially attrited, the 2021 sample could be positively selected, leading estimates to understate true learning losses. Conversely, if pandemic-affected families were more likely to miss testing, estimates could overstate losses. I return to this concern in robustness checks.

Second, most demographic variables show small standardized mean differences (SMDs), suggesting reasonable comparability. Gender composition is nearly identical (SMD = 0.005). Age increased slightly (0.9 months; SMD = 0.161), likely reflecting calendar effects or pandemic-related delays in grade progression. Rural representation increased modestly (25% to 30%; SMD = 0.099).

Third, socioeconomic composition shifted notably. The share of students in the high-SES tertile increased from 21 to 35 percent, while low- and middle-SES shares declined (combined SMD = 0.340). This suggests positive selection on SES in the 2021 sample, reinforcing concerns that estimates may understate losses if disadvantaged students were more likely to attrit. Books-at-home distributions remained broadly similar (SMD = 0.145).

Fourth, school-type composition shifted slightly, with a higher share of gymnasium/lyceum students in 2021 (9% to 13%) and fewer specialized schools (15% to 12%). In sum, while many characteristics remain balanced, the shifts in SES composition and test participation raise valid concerns about selection. The regression analysis controls for observed differences, and I assess sensitivity to compositional change in robustness checks.

Table 3: Sample comparability across waves: Mathematics

Variable	2018	2021	SMD
Sample size			
Enrolled	4,956	4,409	
Tested	4,501	3,781	
Proportion tested	0.908	0.858	
Demographics			
Female	0.497	0.499	0.005
Age (months)	121.5	122.4	0.161
Rural	0.254	0.298	0.099
School type (%)			
Gymnasium/Lyceum	8.7	12.6	
Regular school	76.0	75.4	
Specialized	15.3	12.0	0.150
SES tertile (%)			
Low	30.9	24.1	
Middle	32.2	23.4	
High	21.2	34.5	
Missing	15.6	18.1	0.340
Books at home (%)			
0–10 books	12.4	17.2	
11–25 books	35.1	35.0	
26–100 books	31.5	29.3	
101–200 books	11.9	10.0	
>200 books	7.4	7.0	
Missing	1.6	1.5	0.145

Notes: Weighted means or percentages using survey design weights. SMD = standardized mean difference. Conventional thresholds suggest SMD > 0.10 may indicate imbalance; SMD > 0.25 indicates substantial imbalance.

4 Measuring Learning Loss

4.1 Descriptive evidence

Table 4 reports descriptive statistics on enrollment, test participation, and standardized performance in mathematics and reading for the 2018 and 2021 monitoring cycles. Test scores are standardized relative to the 2018 national distribution. Between 2018 and 2021, average mathematics scores declined by 0.11 standard deviations (SD), while average reading scores declined by 0.10 SD.

To facilitate interpretation, I translate these changes into approximate years of schooling using fourth-grade learning-growth benchmarks proposed by Hill et al. (2008). Under these benchmarks—0.52 SD per school year in mathematics and 0.36 SD per school year in reading—the observed declines correspond to about 0.21 school years of learning loss in mathematics and 0.28 school years in reading. Because these benchmarks are not Ukraine-specific, I treat these conversions as illustrative and emphasize SD units as the primary effect size.

Table 4: Descriptive statistics on enrollment, test participation, and test performance before and after school closures

	2018	2021	Difference
Panel A: Math			
Math score	0.00	-0.11	-0.11
	(1.00)	(1.07)	(0.07)
Proportion tested	0.908	0.858	-0.051
Tested	4501	3781	-720
Enrolled	4956	4409	-547
Panel B: Reading			
Reading score	0.00	-0.10	-0.10
	(1.00)	(1.03)	(0.06)
Proportion tested	0.914	0.857	-0.057
Tested	4506	4143	-363
Enrolled	4928	4832	-96

Notes: Test score rows report weighted means and weighted standard deviations (in parentheses) for tested students. The Difference column reports 2021 minus 2018; for test scores, the standard error of the difference (in parentheses) is obtained from a weighted regression with standard errors clustered at the school level. The proportion tested is tested/enrolled. Final rows report unweighted counts.

Test participation declined between waves, particularly in mathematics, where the proportion of enrolled students who were tested fell from 90.8 percent in 2018 to 85.8 percent in 2021. This motivates explicitly assessing sensitivity to compositional change in the regression analysis below.

Figure 1 plots the weighted distribution of standardized test scores in 2018 and 2021. In both subjects, the 2021 distribution is shifted left relative to 2018, indicating a decline in average

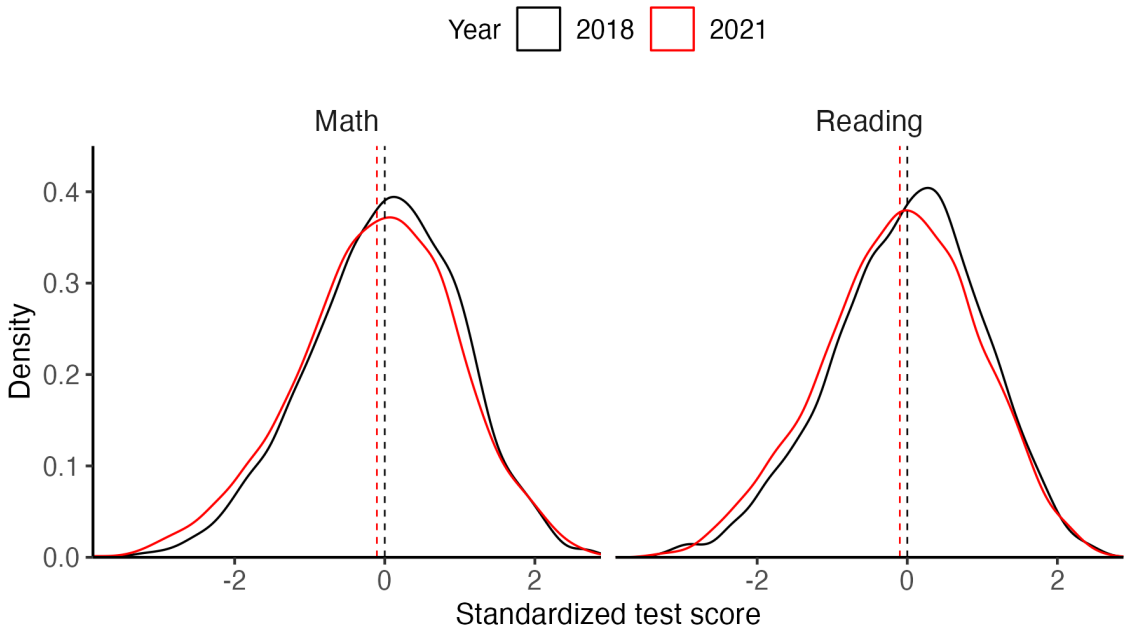

Figure 1: Distribution of standardized test scores in mathematics and reading, 2018 and 2021.
Notes: Weighted kernel density estimates of standardized test scores. Scores are standardized relative to the 2018 national distribution. Dashed vertical lines indicate weighted mean scores in each year.

achievement. The overall shape of the distributions remains broadly similar, suggesting a general shift rather than changes concentrated in specific parts of the distribution.

4.2 Regression specification and interpretation

To estimate cross-cohort changes in achievement, I estimate:

$$Y_{irt} = \alpha_r + \beta_1 \text{May2021}_t + \mathbf{X}'_{irt}\boldsymbol{\beta}_2 + \varepsilon_{irt}, \tag{1}$$

where Y_{irt} is the standardized test score for student i in region r and year t; α_r are region fixed effects; May2021_t is an indicator equal to one for observations from the May 2021 monitoring round (with 2018 as the base period); and $\mathbf{X}_{irt}$ is a vector of student and household characteristics. In my preferred specification, $\mathbf{X}_{irt}$ includes the child's age in months, gender, an indicator for rural residence, and school-type indicators, as well as household background measures such as the number of books at home and socio-economic status (SES). SES is measured using an asset-based index constructed via principal component analysis (PCA), grouped into tertiles (low/middle/high), with a separate category for missing asset information. All regressions use sampling weights and cluster standard errors at the school level.

The coefficient β_1 captures the cross-cohort difference in achievement between 2018 and 2021, conditional on region fixed effects and observed covariates. I interpret this difference as COVID-19–associated under the assumption that, absent the pandemic, the 2021 cohort would have performed similarly to the 2018 cohort after conditioning on observables and region. A

central threat to this interpretation is differential test participation across waves; accordingly, the results section complements Equation (1) with sensitivity analyses that assess robustness to compositional change. I additionally examine heterogeneity by interacting May2021_t with baseline characteristics (e.g., gender, rural residence, and SES).

In Table 5, I estimate cross-cohort differences in standardized achievement between the 2018 (pre-pandemic) and 2021 monitoring waves. Column (1) reports unconditional differences: the 2021 cohort performs about 0.11 SD lower in mathematics and 0.10 SD lower in reading. Adding region fixed effects, school type, and basic student characteristics (column (2)) yields very similar estimates (math: (-0.101) SD; reading: (-0.089) SD), suggesting that differences in regional composition and observable demographics explain little of the raw cohort gap. The preferred specification in column (3), which further controls for household background (books at home and SES tertiles), continues to show a statistically significant decline of roughly 0.09 SD in mathematics and 0.10 SD in reading. Under the maintained assumption that cohorts would have evolved similarly absent the pandemic after conditioning on observables and region, these magnitudes are consistent with modest COVID-19–associated learning shortfalls.

The covariate coefficients exhibit familiar achievement gradients. Rural students score substantially lower than urban students in both subjects (approximately 0.38 SD in mathematics and 0.29 SD in reading in column (3)). Home learning resources are strongly predictive: relative to the lowest books-at-home category (0–10 books), students in households reporting 26–100 books score 0.23 SD higher in mathematics and 0.40 SD higher in reading, with similarly large premia for higher book categories. Missing information on books at home is associated with sizeable deficits in both subjects. Socio-economic status is also positively correlated with achievement, particularly in reading, where students in the high-SES tertile score about 0.12 SD higher than those in the low-SES tertile; missing SES is strongly negatively associated with performance. Finally, gender differences differ by subject: reading shows a large female advantage (0.29 SD), while gender differences in mathematics are small and statistically indistinguishable from zero.

Because the analysis compares repeated cross-sections rather than following the same schools or students over time, the estimates should be interpreted as population-level cohort differences rather than within-school changes. A central threat to the COVID-19 interpretation is differential test participation and compositional change across waves; accordingly, in the next section I assess robustness to alternative reweighting and sample-comparability adjustments and examine heterogeneity in the cohort gap across baseline characteristics.

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Table 5: Cross-cohort differences in achievement: Mathematics and Reading (2018 vs. 2021)

	Mathematics			Reading		
	(1)	(2)	(3)	(1)	(2)	(3)
2021 cohort indicator	-0.105	-0.101*	-0.091*	-0.099*	-0.089*	-0.099**
	(0.067)	(0.056)	(0.050)	(0.059)	(0.046)	(0.040)
School type: Gymnasium / Lyceum		0.233*	0.215**		0.328**	0.253*
		(0.126)	(0.106)		(0.148)	(0.142)
School type: Specialized school		0.050	-0.024		0.096	0.088
		(0.100)	(0.092)		(0.085)	(0.074)
Rural		-0.447***	-0.379***		-0.363***	-0.290***
		(0.091)	(0.082)		(0.089)	(0.077)
Age (months)		0.003	0.004		0.000	0.002
		(0.004)	(0.004)		(0.004)	(0.004)
Female		0.017	-0.012		0.318***	0.288***
		(0.038)	(0.037)		(0.043)	(0.040)
Books at home: 11–25			0.142**			0.233***
			(0.065)			(0.066)
Books at home: 26–100			0.234***			0.403***
			(0.056)			(0.064)
Books at home: 101–200			0.271***			0.382***
			(0.078)			(0.073)
Books at home: >200			0.221**			0.298***
			(0.090)			(0.082)
Books at home: Missing			-0.603***			-0.403***
			(0.140)			(0.152)
SES: Middle tertile			0.091**			0.013
			(0.045)			(0.046)
SES: High tertile			0.067			0.124**
			(0.053)			(0.050)
SES: Missing			-0.609***			-0.578***
			(0.076)			(0.055)
Region fixed effects	No	Yes	Yes	No	Yes	Yes
Controls	No	Yes	Yes	No	Yes	Yes
Observations	8,282	8,128	8,128	8,649	8,578	8,578
R^2	0.000	0.137	0.217	0.000	0.130	0.203

Notes: Each column reports coefficients from survey-weighted regressions of standardized test scores on an indicator for the 2021 cohort (2018 is the reference wave). The dependent variables are standardized using the 2018 weighted mean and standard deviation (separately by subject), so coefficients are in 2018 SD units. Standard errors are design-based and account for survey stratification and clustering at the school (PSU) level. Region fixed effects are included in columns (2)–(3). Column (3) additionally includes books-at-home categories and SES tertiles (with missing categories). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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